

DATA ANALYTICS PROJECT REPORT

Predicting Customer Churn in Digital Banking

A comparative modeling and business insight study, built on 10,000 European bank customers. Exploratory data analysis, feature engineering, comparison of ten classification models, and actionable retention recommendations grounded in the data.

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Executive Summary

Customer attrition is one of the most costly challenges facing retail and digital banks — acquiring a new customer is consistently more expensive than retaining an existing one. This report presents an end-to-end data analytics project built to predict which bank customers are at high risk of churning, understand the behavioural and demographic factors driving that risk, and translate those findings into concrete retention actions.

Using a dataset of 10,000 customers, this study combines rigorous exploratory data analysis with a comparison of ten classification algorithms, hyperparameter optimisation, and a feature importance analysis that is directly cross-validated against the exploratory findings. The final model — a tuned XGBoost classifier — correctly identifies 70% of customers who go on to churn (recall), with an F1-score of 0.63 and 83% overall accuracy, while keeping the pipeline honest about its trade-offs and limitations.

The strongest predictors of churn — number of products held, account activity, age, and geography — align consistently across three independent lines of evidence: visual exploratory analysis, correlation structure, and model-derived feature importance.

MAIN POINT Ten models were compared honestly rather than reporting a single flattering metric. The selected model trades a modest amount of precision for materially higher recall, because in a churn context missing a real churner costs far more than one unnecessary retention offer.

1. Business Problem and Objectives

The Problem

Banks routinely lose customers to competitors without a clear, timely understanding of why. Reactive retention — reaching out only after a customer has already decided to leave — is both less effective and more expensive than proactive intervention. A data-driven early-warning system can help the bank identify at-risk customers while there is still time to act.

Objectives

- Build a classification model that predicts, ahead of time, which customers are likely to churn.
- Identify which customer attributes and behaviours most strongly influence that decision.
- Translate model findings into specific, actionable retention recommendations for the business.

Key Questions

- Which customer segments churn at meaningfully higher rates than others?
- Can churn be predicted reliably enough to be operationally useful, given realistic class imbalance?
- Which single model best balances catching real churners against generating false alarms?

Scope

This analysis is based on a single, static snapshot of 10,000 customer records from a European retail bank, rather than live transactional data. The findings and recommendations should therefore be treated as hypotheses worth validating against real operational data and, where possible, through controlled experimentation (e.g. A/B testing retention interventions) before being deployed at scale.

2. Data and Methodology

Data Source

The analysis uses the publicly available Bank Customer Churn dataset (Kaggle), containing 10,000 records with demographic attributes (age, gender, geography), account characteristics (balance, tenure, number of products, credit score), and engagement indicators (active membership status, satisfaction score). The target variable, `Exited`, indicates whether the customer left the bank (1) or was retained (0). The dataset is imbalanced: only 20.4% of customers churned.

A Note on Data Integrity

The raw dataset includes a `Complain` column, which was deliberately excluded before any modelling took place. This field is recorded in close temporal proximity to a customer's departure and is almost perfectly correlated with churn — including it would constitute data leakage, producing a model that appears highly accurate while relying on information that would not realistically be available at prediction time.

Exploratory Data Analysis Approach

Each of the twelve candidate variables was examined individually against the churn outcome before any modelling began. Categorical variables (`Gender`, `Geography`, `Card Type`, `Number of Products`) were assessed using cross-tabulations and churn-rate bar charts; continuous variables (`Age`, `Balance`, `Credit Score`, `Tenure`, `Estimated Salary`, `Points Earned`) were assessed by comparing group distributions via boxplots. This variable-by-variable approach was deliberately favoured over relying solely on a correlation matrix — a decision that proved important, as detailed in Section 3.

Feature Engineering

Categorical variables were one-hot encoded, with one reference category dropped per variable to avoid the dummy variable trap (perfect multicollinearity). `Gender`, being binary, was mapped directly to 0/1. A derived binary feature, `has_3_or_more_products`, was engineered to explicitly flag the non-linear churn pattern identified for the `NumOfProducts` variable during EDA (see Section 3 and Section 5 for how this feature ultimately performed).

Train / Test Split

Data was split 80/20 into training and test sets using stratified sampling on the target variable, ensuring both sets preserved the original 79.6% / 20.4% class ratio — necessary for a fair, representative evaluation given the class imbalance.

3. Exploratory Findings

Each of the plots below compares the distribution of a single variable, split by churn outcome (`Exited` = 0 retained, `Exited` = 1 churned).

Number of Products — A Non-Linear Signal

Customers holding two products churn least, but churn rises sharply for three products and reaches its highest rate for four. This U-shaped pattern is inconsistent with a simple “more products = more loyalty” narrative, and more plausibly reflects aggressive cross-selling: customers pushed into products they neither needed nor wanted.

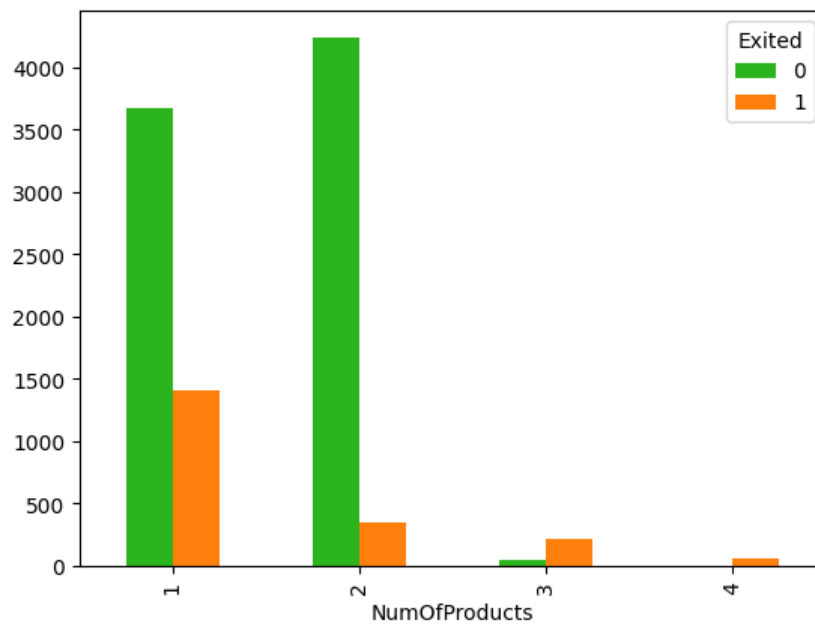


FIGURE 1 — Customer count by number of products held

Customers split by churn outcome across each product-count category.

Account Activity and Geography

Inactive members churn at nearly double the rate of active members, consistent with the intuition that weaker engagement precedes disengagement. Geography also shows a pronounced effect: German customers churn at roughly double the rate of French and Spanish customers, though the underlying cause cannot be determined from this dataset alone (see Recommendation 4).

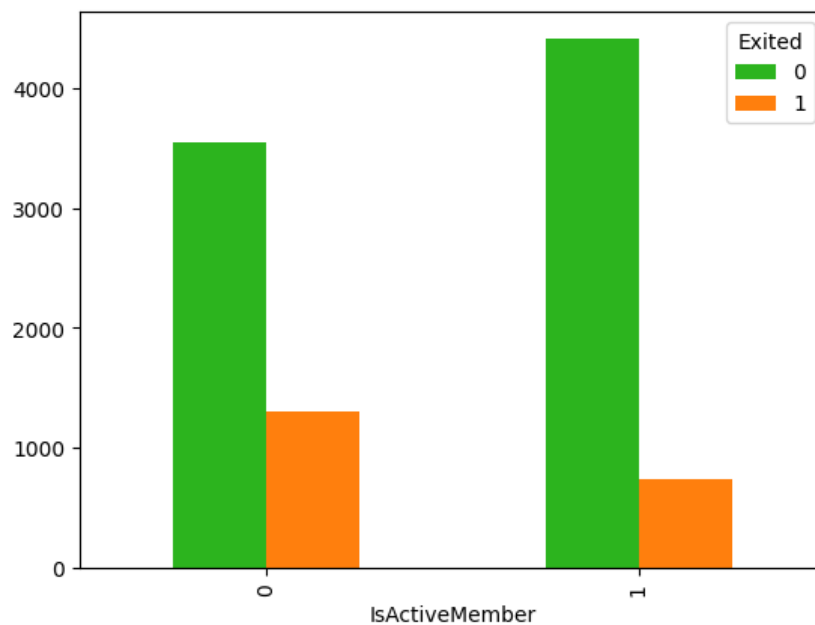


FIGURE 2 — Customer count by membership activity

Active vs. inactive members, split by churn outcome.

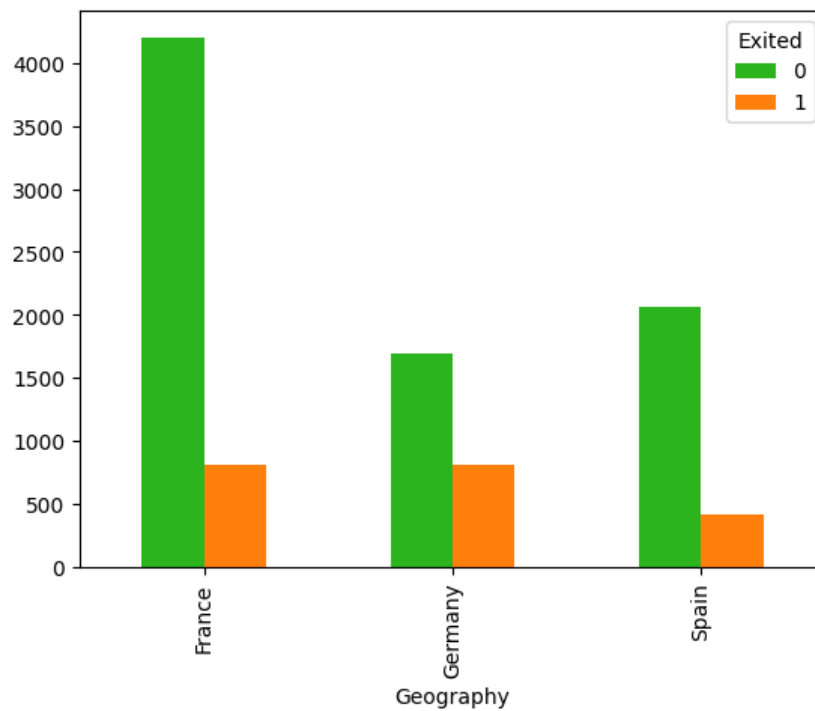


FIGURE 3 — Customer count by geography

France, Germany, and Spain, split by churn outcome.

Age and Gender

Retained customers cluster in their 30s, while churned customers cluster in their 40s to early 50s — a moderate but consistent signal. Female customers churn more often than male customers, a moderate effect worth further investigation.

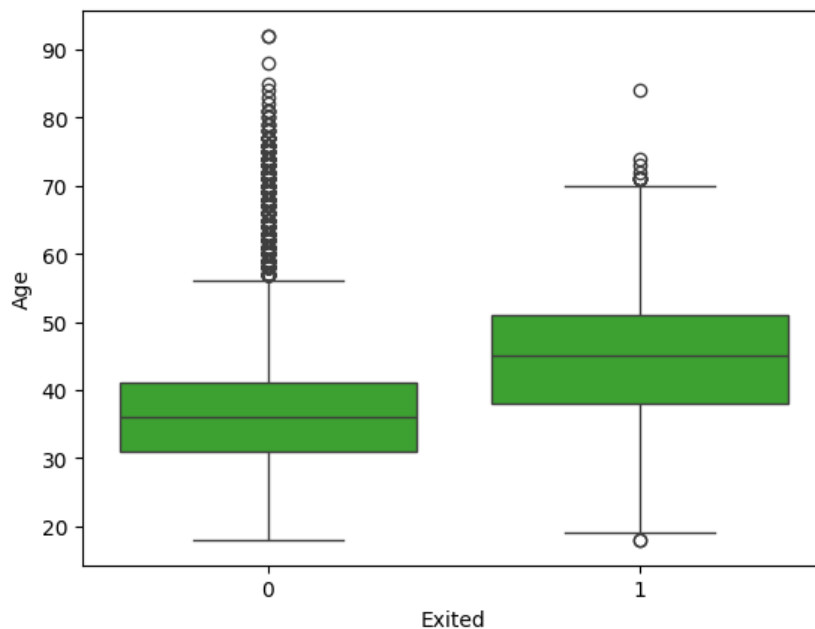


FIGURE 4 — Age distribution by churn outcome

Retained customers skew younger than churned customers.

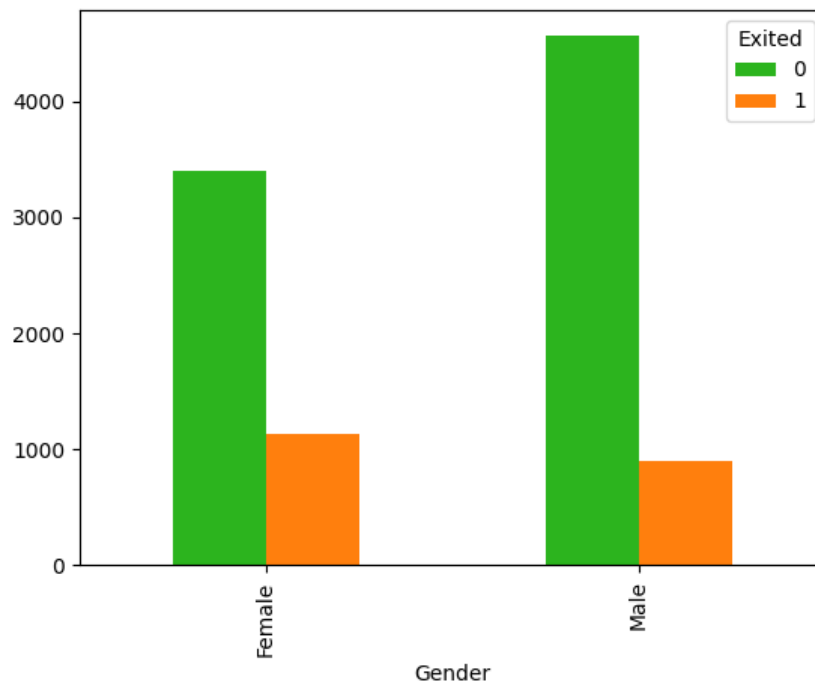
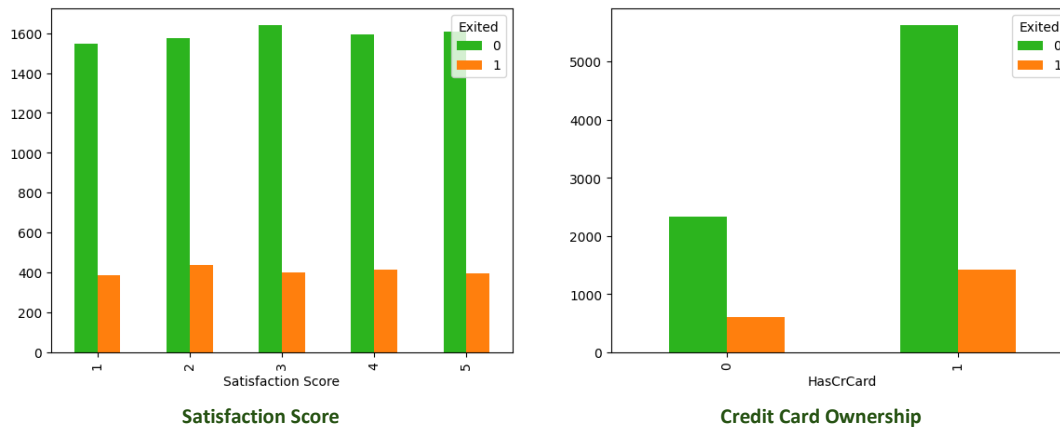


FIGURE 5 — Customer count by gender
Split by churn outcome.

Variables With Weak or No Signal

Balance, Credit Score, Estimated Salary, Card Type, Satisfaction Score, and Points Earned showed little to no meaningful difference between churned and retained customers. Notably, self-reported Satisfaction Score carried almost no predictive signal — a counter-intuitive finding suggesting that stated satisfaction does not reliably predict actual retention behaviour.



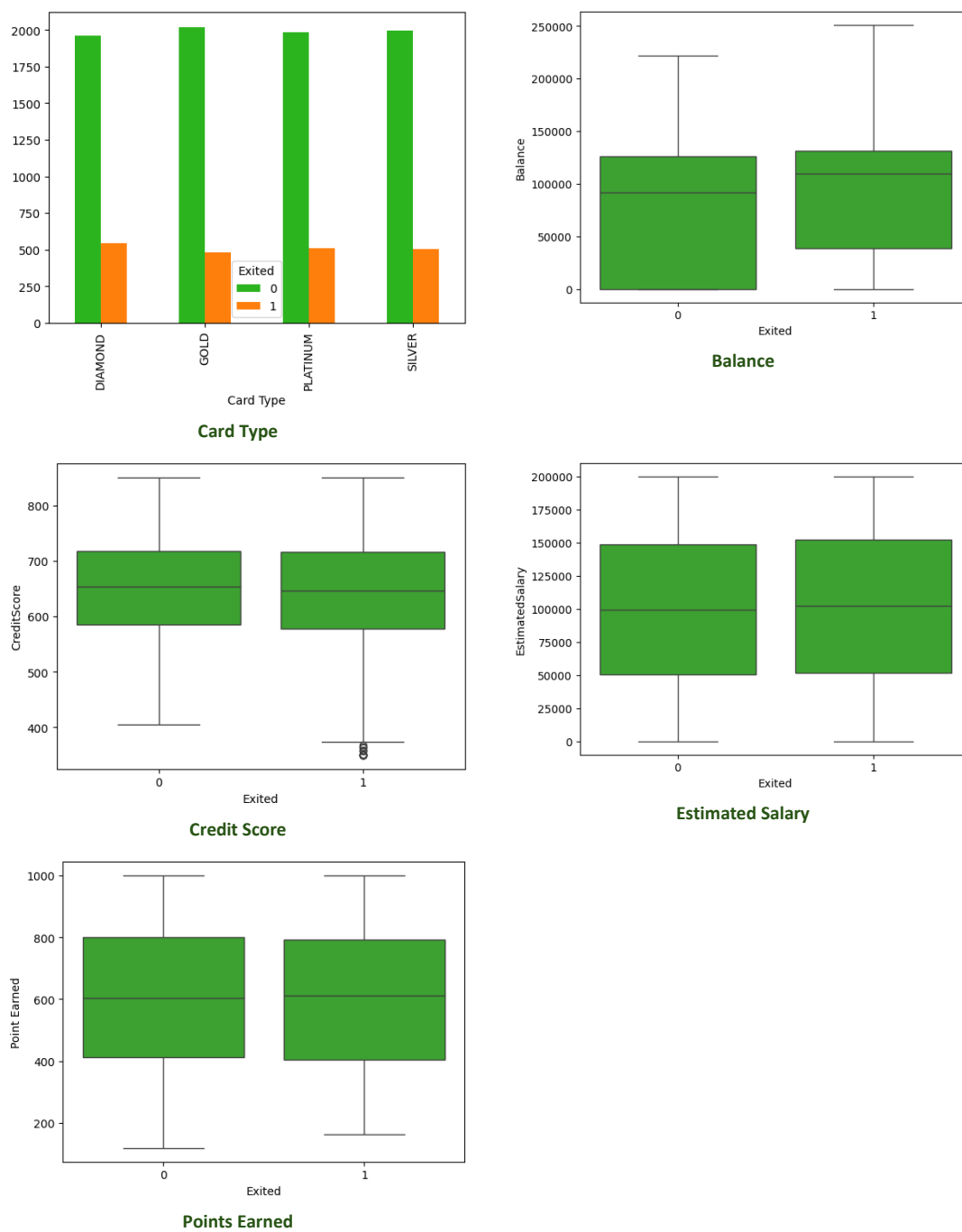


FIGURE 6 — Weak-signal variable distributions

Each panel split by churn outcome; none shows a meaningful separation.

Correlation Structure and Its Limits

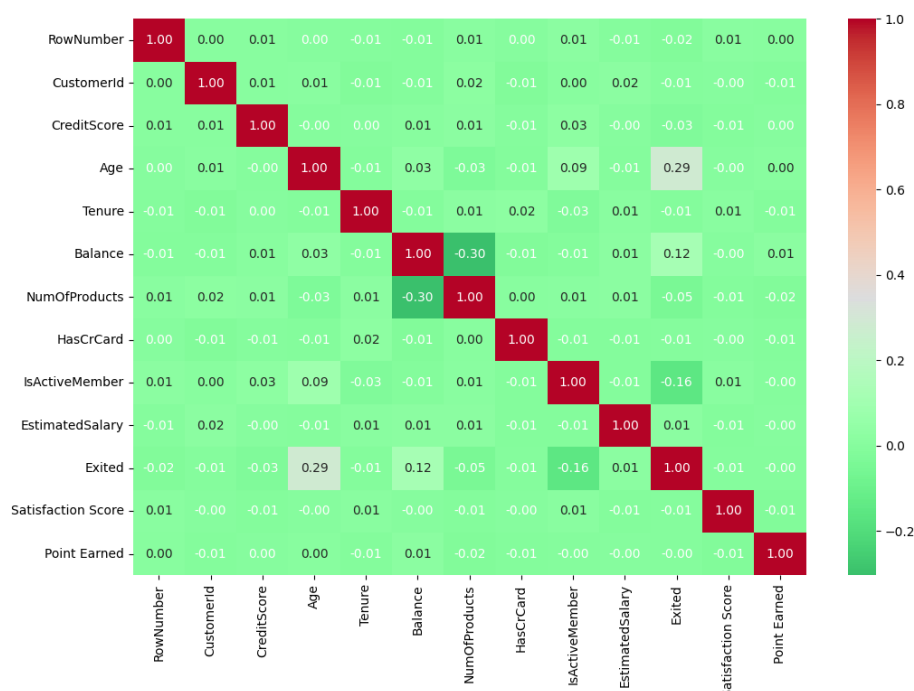


FIGURE 7 — Pearson correlation matrix across all numerical variables

The correlation matrix confirms Age (0.29), IsActiveMember (-0.16), Balance (0.12), and Gender (0.11) as the variables with the strongest linear relationship to churn. Critically, however, NumOfProducts — the single strongest predictor identified through visual EDA — shows almost no linear correlation with churn (-0.05).

Pearson correlation only captures linear relationships, and the NumOfProducts effect is U-shaped, so its upward and downward segments partially cancel out mathematically. This directly justifies the decision to conduct variable-by-variable visual EDA rather than relying on correlation analysis alone — a methodological choice later validated by the model's own feature importance ranking (Section 5).

4. Modeling

Handling Class Imbalance

With only 20.4% of customers churning, a naive model can achieve ~80% accuracy by predicting “no churn” for everyone — a useless outcome in practice. Two imbalance-handling strategies were tested depending on model type: `class_weight='balanced'` for Logistic Regression, Random Forest, and SVM, and `scale_pos_weight` for XGBoost. Each was benchmarked against its unweighted counterpart rather than assumed to help by default.

Models Compared

Ten classification models were trained and evaluated on the same held-out test set: Logistic Regression (original and class-weighted), Random Forest (default and class-weighted), XGBoost (default, class-weighted, and hyperparameter-tuned via GridSearchCV), Support Vector Machine, Gradient Boosting, and Gaussian Naive Bayes. Numerical features were standardised for the scale-sensitive models; tree-based models used unscaled features.

Hyperparameter Optimisation

GridSearchCV with 3-fold cross-validation was used to tune XGBoost across a grid of max_depth, learning_rate, and n_estimators, optimising for F1-score. The best configuration (max_depth=7, learning_rate=0.1, n_estimators=100) improved F1 from 0.60 (class-weighted default) to 0.63 and Recall from 0.64 to 0.70 — a meaningful, if modest, gain consistent with the expectation that tuning typically yields incremental rather than dramatic improvements on a dataset of this size and structure.

Model Selection Rationale

The tuned XGBoost model was selected as the final model, prioritising Recall alongside F1-score. In a churn context, the cost of a false negative (missing a customer who is about to leave) substantially outweighs the cost of a false positive (a retention offer sent to someone who was never going to leave). Random Forest (Default) and Gradient Boosting achieved slightly higher accuracy and precision, but at a meaningfully lower recall — a trade-off less favourable for this specific business objective.

A Note on Naive Bayes

Gaussian Naive Bayes performed dramatically worse than every other model (F1: 0.11, Recall: 6%). This is consistent with its core assumption of feature independence, which is clearly violated in this dataset — for example, Balance and NumOfProducts are correlated at -0.30. Rather than a failed experiment, this result serves as an independent confirmation, from a modelling angle, of a structural property already observed in the correlation analysis.

5. Results

Performance Comparison

Model	Accuracy	Precision	Recall	F1
XGBoost (Tuned)	0.83	0.57	0.70	0.63
Gradient Boosting	0.87	0.78	0.51	0.615
Random Forest (Default)	0.87	0.80	0.48	0.60
XGBoost (Balanced)	0.83	0.57	0.64	0.60
SVM (Balanced)	0.78	0.48	0.75	0.58
Logistic Regression (Balanced)	0.77	0.46	0.77	0.58
XGBoost (Default)	0.85	0.68	0.50	0.58
Random Forest (Balanced)	0.86	0.80	0.45	0.57
Logistic Regression (Original)	0.84	0.71	0.37	0.48
Naive Bayes	0.79	0.37	0.06	0.11

TABLE 1 — All ten models, ranked by F1-score on the held-out test set.

Confusion Matrix — Final Model

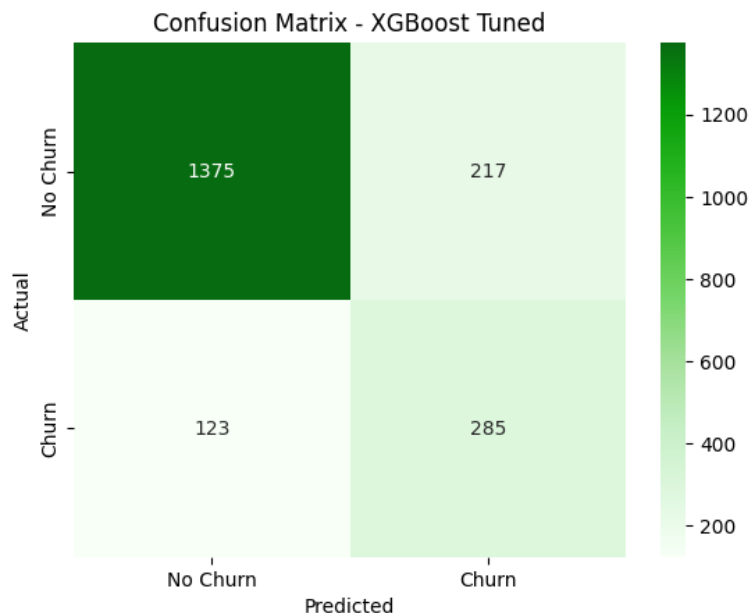


FIGURE 8 — Confusion matrix for the tuned XGBoost model

Test set, n = 2,000.

Of the 408 customers who actually churned in the test set, the model correctly flagged 285 (70%), while 123 went undetected. It also flagged 217 customers as high-risk who ultimately stayed. Given that the cost of an unnecessary retention offer is typically far lower than the cost of losing a customer with no intervention, this trade-off is favourable from a business standpoint.

Feature Importance

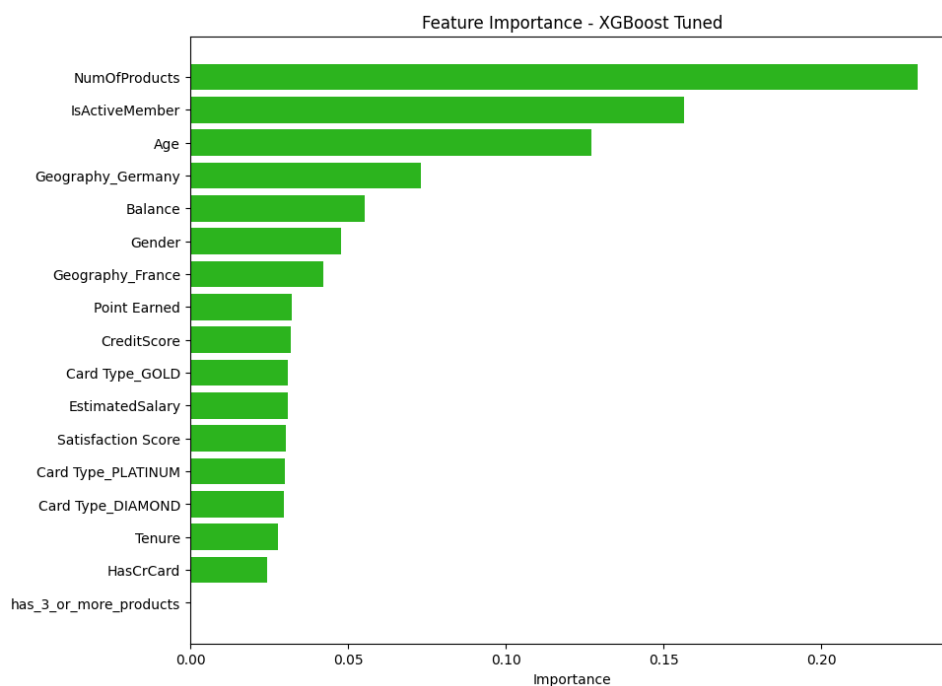


FIGURE 9 — Gain-based feature importance from the final tuned XGBoost model

Feature importance from the final model closely mirrors the exploratory findings: NumOfProducts, IsActiveMember, Age, and Geography_Germany rank as the top four predictors — the same variables identified as the strongest signals through visual EDA in Section 3. This convergence across two independent analytical approaches substantially strengthens confidence in these findings.

The engineered feature `has_3_or_more_products` received zero importance. XGBoost, as a tree-based non-linear model, already captures the U-shaped NumOfProducts pattern directly from the original variable — rendering the derived binary feature redundant here. It would likely have added value for a linear model such as Logistic Regression, which cannot natively represent non-linear relationships without explicit feature engineering: feature engineering value is model-dependent, not universal.

6. Business Recommendations

The following recommendations translate the strongest predictive signals into concrete, actionable retention strategies.

1. Reconsider cross-selling strategy beyond two products

Aggressive upselling beyond two products correlates with sharply higher churn rather than increased loyalty. Recommend proactive account check-ins — framed around service quality rather than additional sales — for customers who have recently acquired a third or fourth product.

2. Launch re-engagement campaigns for inactive members

Inactivity is one of the strongest churn predictors. Recommend early intervention after 30–60 days of inactivity, using personalised notifications or small incentives to resume regular account activity before disengagement becomes permanent.

3. Review product-market fit for customers aged 40–55

This age group churns more than customers in their 30s and may have different financial priorities — mortgages, retirement planning, or wealth management. Recommend evaluating whether current offerings skew toward younger, digital-first customers and consider tailored advisory services for this segment.

4. Investigate the German market specifically

Churn in Germany is roughly double that of France and Spain, but the dataset does not reveal the underlying cause. Recommend a market-specific survey or focus group to determine whether local competition, product fit, or service quality is the primary driver, before designing a targeted intervention.

7. Limitations and Future Work

Static, single-snapshot data

The dataset represents a single point in time with no transactional history, meaning the model cannot capture trends in customer behaviour leading up to churn. Incorporating time-series behavioural features would likely improve predictive power substantially.

Unverified causal hypotheses

Several recommendations (e.g. the German market effect, the age-related product gap) are grounded in observed statistical patterns but not in confirmed causal mechanisms. These should be treated as hypotheses to validate through further qualitative research or controlled experimentation, not as established facts.

Recall–precision trade-off

The selected model prioritises Recall, accepting a meaningful number of false positives (217 in the test set). Depending on the actual cost of a retention offer versus the value of a retained customer, the decision threshold could be further tuned to shift this balance.

Opportunities for further modelling

Future iterations could explore SMOTE-based oversampling as an alternative to class weighting, probability calibration to better estimate individual churn risk, and cost-sensitive threshold tuning aligned with the bank's actual retention budget.

8. Conclusion

This project demonstrates a complete, honest, and reproducible approach to churn prediction: from rigorous exploratory analysis through to a rigorously benchmarked final model. Rather than reporting a single flattering metric, this report presents ten models side by side, explains why the selected model is the right choice for the business objective — not simply the one with the highest accuracy — and grounds every recommendation directly in evidence that appears consistently across independent analytical methods.

The resulting model, a tuned XGBoost classifier achieving 70% recall and an F1-score of 0.63, provides the bank with a practical, immediately usable tool to prioritise retention efforts — while the underlying analysis surfaces four concrete, testable interventions that go well beyond the model itself.

PORTFOLIO STATEMENT This project demonstrates the ability to run a complete analytics workflow end to end: rigorous EDA, honest model comparison, and business recommendations grounded in reproducible evidence rather than a single flattering metric.

Full code, notebook, and dataset documentation available at: github.com/todorovalex/bank-customer-churn-prediction